

F24-126-D-FinXtract

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Chapter 1

Introduction

Our project FinXtract is a web-based software application which is based on automating the extraction and analysis of data from financial reports. This product aims to solve the problems that arise due to the manual extraction of data by completely automating this process, reducing the need for human intervention. The software includes features such as data standardization which is a significant need to ensure accurate comparison between companies and a chatbot designed for financial data query assistance as well as providing sentiment analysis and summarization of reports to evaluate performance of a company. By eliminating the need for manual extraction, FinXtract helps to reduce the errors and time taken during data extraction, providing accurate and reliable data for efficient financial analysis. The requirements and specifications of FinXtract are described in this document. The document is intended for a wide range of readers who will be affected by the product's development and use. The following readers will use this document for different tasks.

- **Users**, which will include Financial analysts and Investors
- **Developers**, who will build and maintain the product.
- **Project Managers**, who will manage the timelines and resources.
- **Marketing Staff**, who will promote the product to expand its reach.
- **Testers**, who will ensure quality and high performance.
- **Documentation Writers**, who will create user guides and manuals.

1.1 Problem Statement

In today's financial world, companies rely on employees who read whole PDF reports of banks to gather significant financial data. This whole process is done manually. While this can also take a lot of time since this task is performed by humans, there is still no guarantee that the data gathered will be fully accurate. There may be mistakes as a result of human error. So, we are developing our product FinXtract with the main goal of automating this process of financial data extraction from PDFs. This will eliminate the possibility of human error and make this process as accurate as possible. This will enable researchers and analysts to fully focus on their objectives respectively rather than on how accurate

the extracted data is. Additionally, different companies use different variables to define their financial terminologies, for example, one company might use the word sales while another might describe it as revenue. As a result, there is high variation in the data (which can also occur in the different reports of the same company). This high variation can lead to inadequate data analysis. Our product will be able to tackle this issue through standardization and help companies make informed and data-driven decisions.

1.2 Scope

The purpose of our project is to develop a software tool that will simplify and speed up the process of both data extraction and analysis from financial reports in PDF format. It will automatically retrieve data from both regular and scanned PDFs, reducing manual work and saving time. The tool will be user-friendly, making it easy for financial analysts, investors, and others to access the data they need. FinXtract will include features like sentiment analysis to assess company reports, helping users understand business performance and trends. It will also have an intelligent chatbot for real-time help with queries and data retrieval, ensuring accuracy and security. The system will be able to securely store all extracted data and ensure it is standardized and cleaned for accurate analysis. The project will focus on accuracy and consistency to ensure reliable results. Overall, FinXtract is designed to make financial data extraction and analysis easier, more accurate, and secure.

1.3 Modules

1.3.1 Module 1: Data Scraping

This module ensures that the process of downloading financial PDF reports from company websites is automated.

1. **Automated PDF Downloading:** Automates the process of downloading PDFs of various companies from their websites and storing them in their respective folders.
2. **Multi-Level Scraping:** Involves downloading PDF reports either from the main page (one-level) or by navigating through sub-pages (two or three-level).

1.3.2 Module 2: Financial Data Extraction

This module ensures the extraction and processing of financial data from PDFs once they have been downloaded for further analysis.

1. **Data Extraction through OCR:** Utilizes Optical Character Recognition to extract tabular and textual data from scanned or image-based PDF reports.

1.3.3 Module 3: Standardization

This module ensures that the extracted financial data is standardized for consistency and accurate data analysis.

1. **Reconstruction of Elements:** Reconstructs the extracted tabular data into sentences, converting it from unstructured to structured format.
2. **Data Consistency:** Ensures that the data extracted from PDFs of various companies or the same company across different years have the same format used.

1.3.4 Module 4: Chatbot

This module ensures the creation of an interactive chatbot that helps in the retrieval of data by generating answers to user queries.

1. **Data Querying:** Allows users to answer questions regarding specific financial information based on their needs.
2. **NLP:** Uses NLP to generate summarization and sentiment analysis of PDF reports. Sentiment Analysis will help to identify the positive, negative, and neutral sentiments across different reports over time.

1.3.5 Module 5: Front-end Integration

This module ensures that the financial data is displayed on the front end along with the chatbot.

1. **Data Visualization:** Displays Reported and Standardized data of various companies across different years in the form of tables.
2. **ChatBot Integration:** Creates a user-friendly chatbot design to handle user queries.

1.4 User Classes and Characteristics

Table 1.1: User Classes and Characteristics

User class	Description
Financial Analysts	Financial analysts are professionals who play a very important role in the understanding of financial statements, market trends, and company performance to provide investment advice or recommendations. They have to deal with large volumes of data and need to perform in-depth analyses under tight deadlines. Due to the complexity and critical nature of their work, they require tools that simplify data extraction and analysis, so that they can concentrate more on generating analysis reports rather than focusing on manual data gathering.
Investors	Investors are individuals who rely heavily on the accurate and timely availability of financial data so they can make informed decisions about capital allocation. They need to properly analyze the company performance and the risk related to help them guide their investment strategies effectively. Efficient analysis of critical data is of greatest importance for them, as they often need to act swiftly in a competitive investment landscape.
Accountants and Auditors	Accountants and auditors make sure of the accuracy and compliance of financial documents. They carefully review financial statements to find inconsistencies and verify data integrity. Their work requires a strong understanding of regulatory standards and financial reporting. They operate under very strict deadlines, particularly when conducting audits, which is why they rely on accurate data extraction tools to deliver authentic reports.
Corporate Executives (CFOs, CEOs)	Corporate executives which include CFOs and CEOs are the backbone of a company who are responsible for making important company decisions. To make crucial decisions, they need quick access to significant financial information and clear company performance summaries. They seek high-level insights to guide company objectives rather than focusing on specific data. Effective planning for strategies essentially requires reliable access to financial data because their responsibilities include team management and stakeholder communication.
Regulators and Compliance Officers	Regulators and compliance officers, by working in governmental organizations ensure that businesses abide by financial laws and guidelines. Their focus relies mainly on the details of the financial reports to look for any irregularities in the data to prevent fraud. Precise and consistent data analysis is a necessary requirement for them to satisfy legal requirements. Accurate and standardized data extraction technologies can help them accelerate their review procedures and monitor compliance among companies more effectively.
Financial Consultants	Financial consultants advise companies on investment decisions, planning and enhancing their financial condition. They compare several financial reports to offer specific recommendations because they possess strong analytical abilities and industry understanding. For their analysis, timely access to reliable financial data is essential. Consultants can provide better guidance and assist clients in making informed financial decisions by utilizing effective data extraction tools.

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

2.1.1 Use Case Diagram

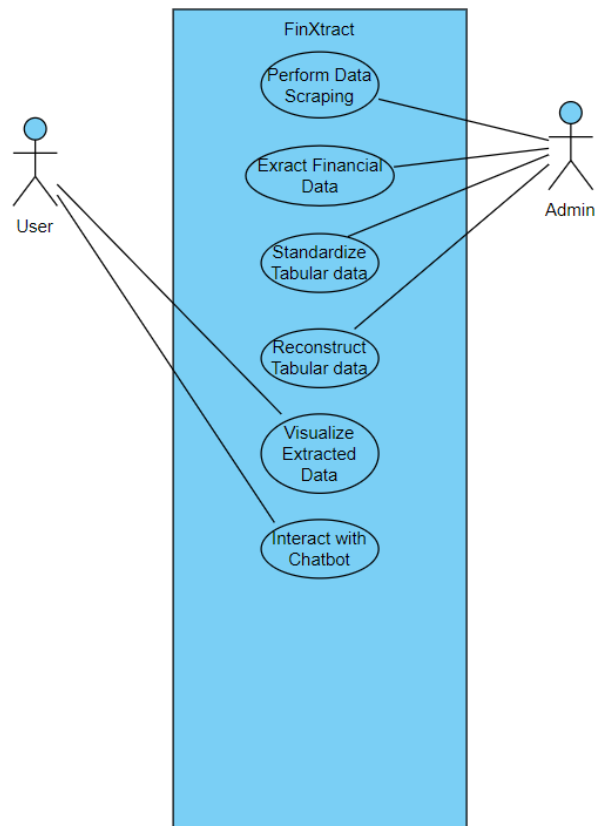


Figure 2.1: Use Case Diagram

2.1.2 Event Response Table

Table 2.1: Event and System Response

Event	System Response
The system scrapes a website for financial data analysis	The data scraping module finds the PDFs on the company website and downloads them to the system's storage, following multi-level navigation if needed.
OCR processing of scanned PDFs	OCR converts the image-based text into machine-readable text, extracts both tabular and unstructured text, and passes it to the sentence reconstruction and standardization module.
Data standardization is triggered	The standardization module cleans the data and converts it into a uniform format, correcting differences like "sales" vs. "revenue".
Sentiment Analysis Request	The system processes the financial data and provides the sentiment analysis results.

2.2 Functional Requirements

2.2.1 Module 1: Data Scraping

1. **FR1:** The system shall download PDF reports automatically from banking company websites without any intervention.
2. **FR2:** The system shall support multi-level scraping, that allows for the downloading of PDF reports from main pages as well as sub-pages.

2.2.2 Module 2: Financial Data Extraction

1. **FR1:** The system with the help of OCR shall be able to Extract Tabular and Textual Data from Scanned / Image-based PDF Report(s)
2. **FR2:** The system shall be able to extract data from the multiple pages of a PDF document, helping to make sure that all necessary accounting figures existing across various sections are captured.

2.2.3 Module 3: Standardization

1. **FR1:** The system shall convert the raw unstructured data into a structured format for easier analysis.
2. **FR2:** The system shall be able to standardize the financial data extracted from different companies or periods to make it consistent.

2.2.4 Module 4: Chatbot

1. **FR1:** The system shall give real-time answers of user queries related to financial data using the help of a chatbot.
2. **FR2:** The system shall process financial reports by summarizing and carrying out sentiment analysis with the help of a natural language processing (NLP) module.

2.2.5 Module 5: Front-end Integration

1. **FR1:** The system shall display As-Reported and As-Standardized extracted financial data in user-friendly tables on the front end.
2. **FR2:** The system shall integrate the chatbot into the user interface for easy access to information related to financial reports.

2.2.6 Module 6: Testing and Deployment

1. **FR1:** The system must be tested extensively before deployment so that all its features work in the way that they should.
2. **FR2:** The system shall be user friendly and easily accessible to users in post-deployment mode.

2.3 Non-Functional Requirements

2.3.1 Reliability

REL-1: The system shall be highly reliable, minimizing downtime, and achieving an MTBF of at least 720 hours, ensuring that failures are rare.

REL-2: The system shall recover from failures such as server crashes within 2 minutes to ensure optimal service to users.

REL-3: The system shall provide clear error messages and recover from uploading Unsupported document formats without crashing.

2.3.2 Usability

USE-1: The system shall be user-friendly, enabling new users to perform basic tasks without the need to learn about the system from someone.

USE-2: The system shall allow the user to view previously extracted data from financial reports within 2 clicks, ensuring ease of use and avoiding unnecessary interactions.

USE-3: The system shall provide clear error messages, guiding the user in case of uploading an unsupported file type for data extraction.

2.3.3 Performance

PER-1: The system shall respond to user inputs including data access and navigation in no more than 5 seconds.

PER-2: The system shall be able to process large datasets efficiently without a significant drop in performance.

PER-3: The system shall ensure that interactions with the backend, such as data queries and retrieval are processed with minimal delay.

2.3.4 Security

SEC-1: The system shall be designed to prevent unauthorized access, requiring a high amount of skill to gain entry into the system.

SEC-2: The system shall ensure data integrity during storage and transmission.

SEC-3: The system shall protect sensitive financial data at all times, making any attempt to misuse the data extremely difficult.

2.3.5 Scalability

SCA-1: The system shall be designed for deployment in a cloud environment using services such as AWS for easy access.

SCA-2: The system shall efficiently scale to handle increased number of users without a drop in performance.

2.4 Domain Model

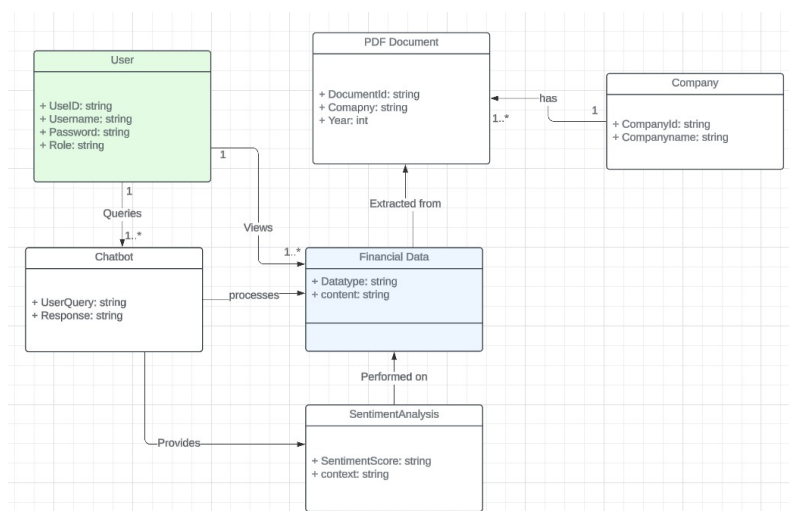


Figure 2.2: Domain Model

Chapter 3

System Overview

FinXtract is a software tool that automates extracting and analyzing financial data from PDF reports, saving time and improving accuracy. It helps analysts, investors, auditors, and executives to quickly access clean and standardized financial information.

FinXtract does the job of extracting and standardizing financial data from PDFs, thus ensuring that all of its users have access to clean and ready-to-use information. It uses OCR for extracting text from scanned documents, NLP for performing Sentiment Analysis, and complex algorithms for standardizing different terms like "sales", and "revenue" used by the companies. FinXtract also includes a smart chatbot that can be of great use for providing real-time assistance to users, answering their queries, and fetching important information from reports.

Companies, analysts, and auditors often spend a significant amount of their time going through numerous financial documents manually to retrieve information, which can lead to errors. The fact that companies have used different terms over time only further complicates data comparison. The use of scanned documents also doesn't simplify the process of information retrieval.

FinXtract overcomes all these problems by automating data extraction, ensuring consistency and accuracy of the financial data. Its easy-to-use interface allows professionals to focus more on analysis and decision-making rather than worrying about the accuracy of the extracted data.

It is a cloud application for pulling data from PDFs. Its key modules are:

- **PDF Processing:** Extracts text from regular and scanned PDFs using OCR.
- **Data Standardization:** Maintains consistency in financial terms.
- **Sentiment Analysis:** Analyzes the performance of the companies over time.
- **Chatbot:** Provides real-time answers to user queries.
- **User Interface:** A web-based interface for easy access to the reported and standardized data and a chatbot for better user experience.

3.1 Architectural Design

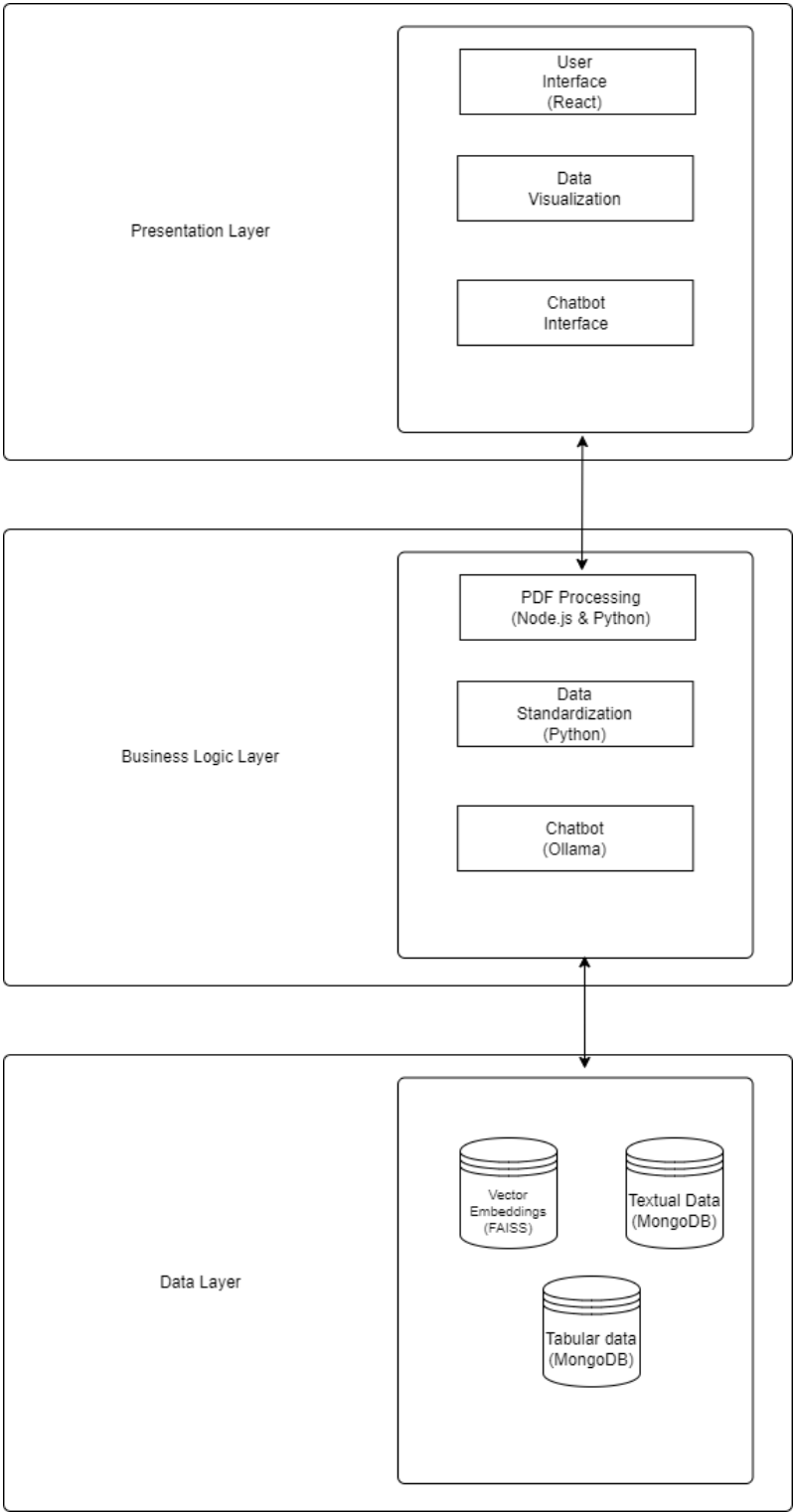


Figure 3.1: Architecture Diagram

FinXtract's system architecture is a modular architecture that helps to make the system very flexible, scalable and easy to maintain. Most of the modules have some sort of functionality or design associated with specific tasks that help to reach the main goal of extraction and analysis of financial data. All these modules are automatically linked together to cover the entire workflow from scraping financial PDFs, to standardizing data and showing them in a user-friendly interface.

3.1.1 High-Level Design Overview

FinXtract uses a Layered Architecture model, which divides the system into separate, independent layers for better organization:

1. **Presentation Layer (Front-end):** It consists of a User Interface (UI) that shows the data scraped through PDFs in the form of reported and standardized data and a chatbot. This user-friendly interface allows the users to use the system conveniently according to their needs.
2. **Business Logic Layer:**
 - This layer consists of primary modules that manage data extraction, standardization, and analysis.
 - **PDF Processing Module:** Converts raw data from PDF reports into structured formats.
 - **Data Standardization Module:** Ensures that the data of multiple companies is consistent and ready for analysis.
 - **Chatbot Module:** Responds to user questions and provides real-time assistance using natural language processing.
3. **Data Layer:** This layer securely stores all the processed and standardized data, making it easily accessible for analysis and chatbot responses.

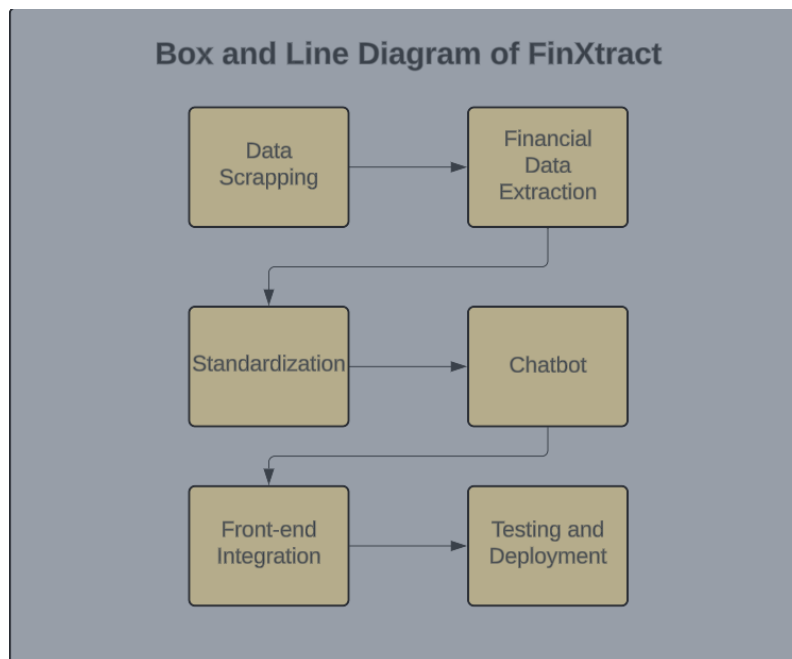


Figure 3.2: Box and Line Diagram

3.1.2 Module Interactions

1. **Data Scraping Module:** Retrieves PDFs from company websites and sends them to the Financial Data Extraction Module for processing.
2. **Financial Data Extraction Module:** Processes both regular and scanned PDFs, converting the data into structured formats for the Standardization Module.
3. **Standardization Module:** Ensures consistency by making different financial terms the same, allowing for easy comparison between companies and years.
4. **Chatbot Module:** This module uses standardized data and natural language processing techniques to help the user in the retrieval of queries and report summaries.
5. **Front-end Integration Module:** Displays the extracted reported and standardized data, allowing users to access both types of data and chatbot responses.

3.2 Design Models

3.2.1 Activity Diagrams

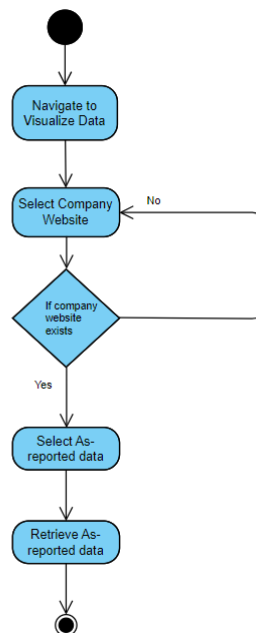


Figure 3.3: Visualize As-reported Data

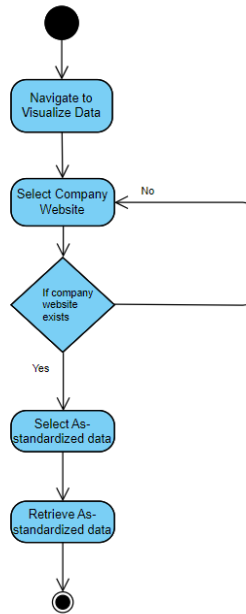


Figure 3.4: Visualize As-standardized Data

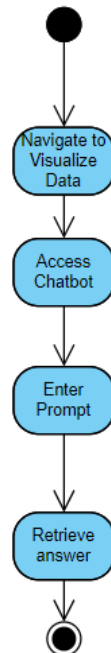


Figure 3.5: Chatbot Interaction

3.2.2 Data Flow Diagram

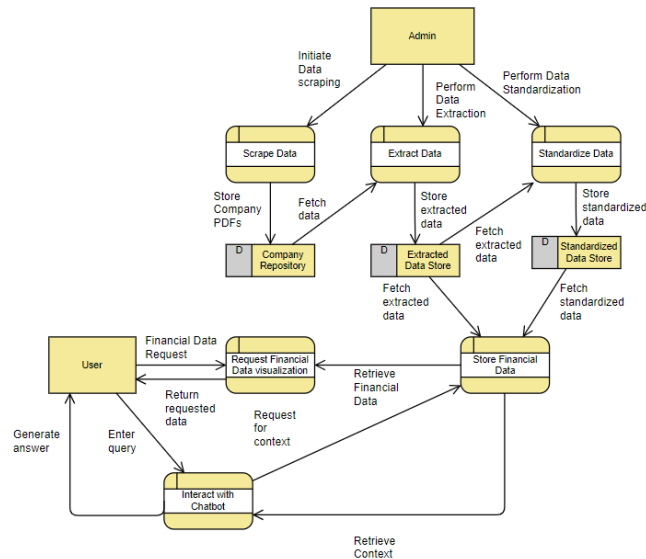


Figure 3.6: Data Flow Diagram

3.2.3 System Sequence Diagram

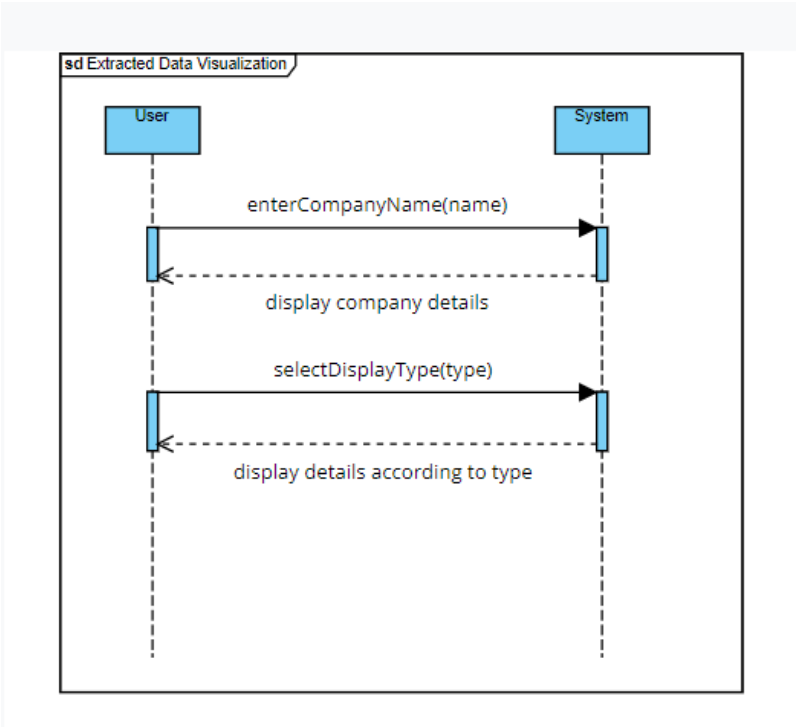


Figure 3.7: Visualize Extracted Data

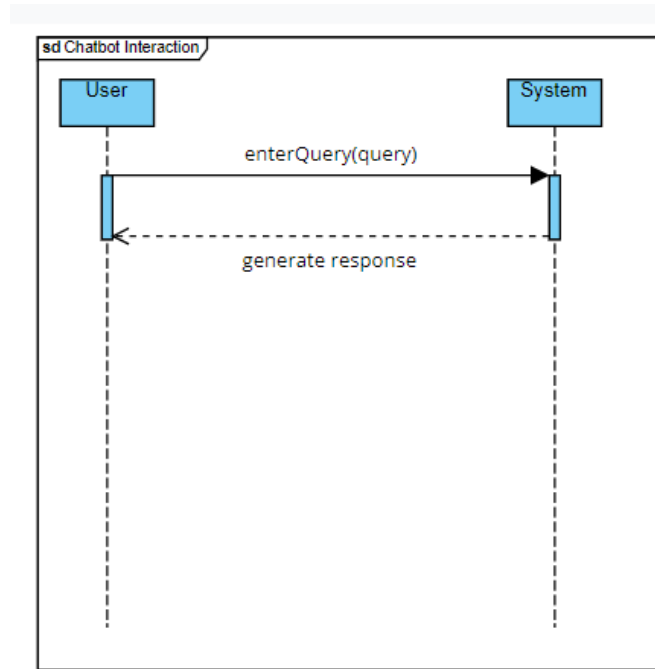


Figure 3.8: Interact with Chatbot

3.2.4 State Transition Diagram

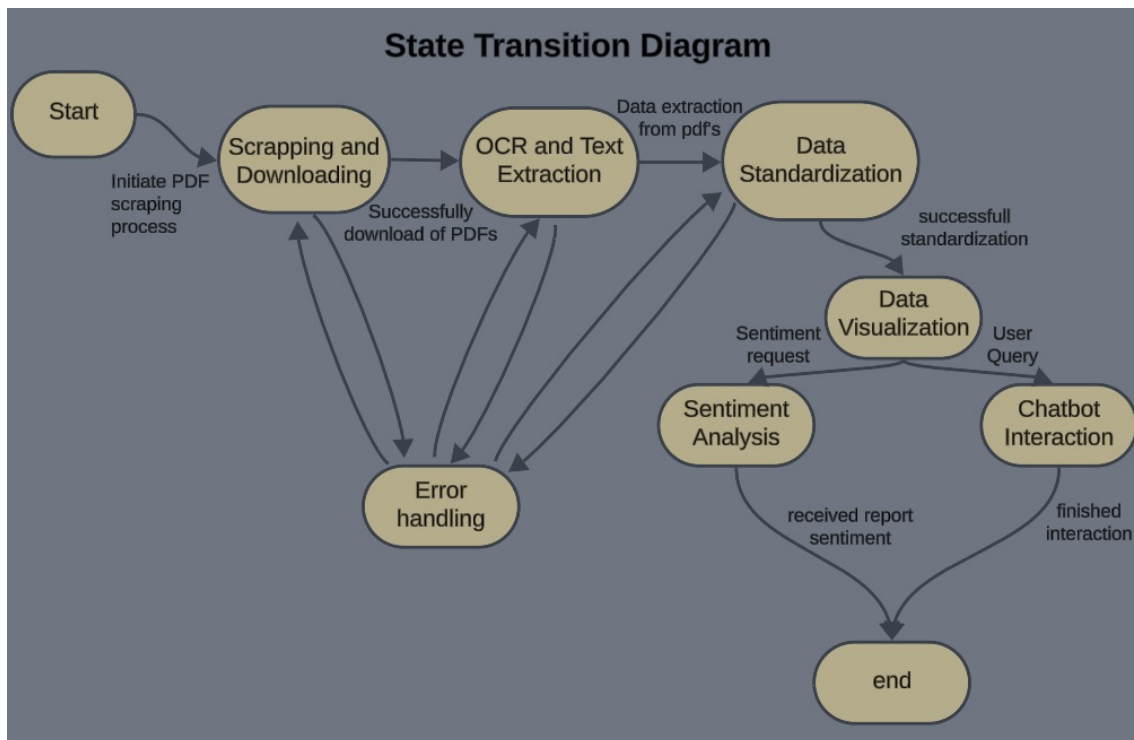


Figure 3.9: State Transition Diagram

3.2.5 ERD

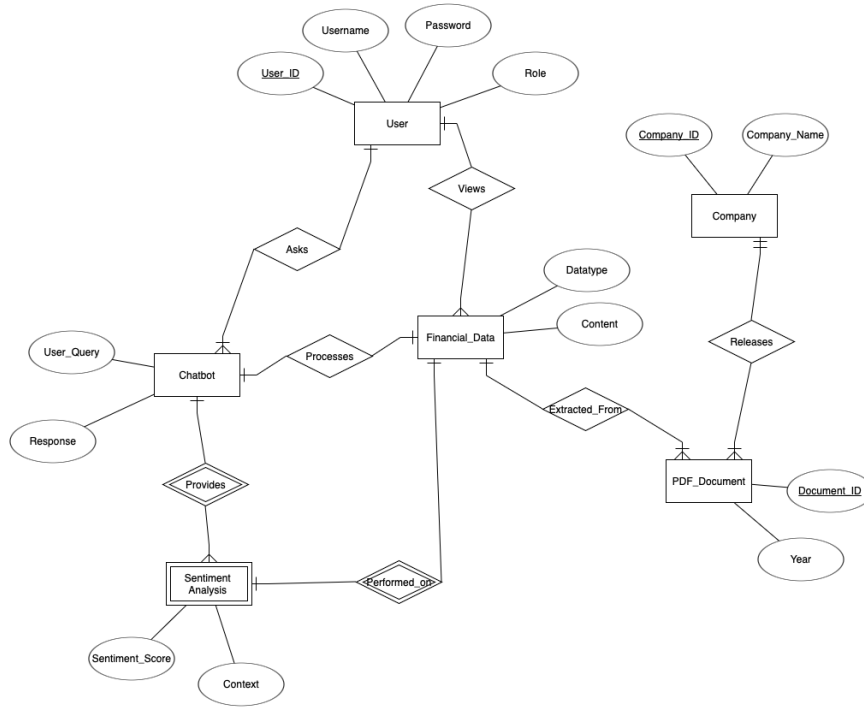


Figure 3.10: ERD

The ERD Diagram of FinXtract includes entities such as a User who will be interacting with the website after entering his/her credentials. There are various companies who have financial PDF documents from whom the financial data is extracted. The User views the extracted data (reported or standardized) according to their needs as well as interacts with the chatbot. The chatbot processes financial data when generating an answer and provides sentiment analysis which is a weak entity in our architecture since sentiment analysis cannot exist without a chatbot. The sentiment analysis is performed on financial data which is also a weak relationship due to the dependency of sentiment analysis on the chatbot.

The UserID is primary for each user since each user is unique. Same is the case regarding CompanyID for a company and DocumentID for a PDF document. A single company releases many pdf documents (1-Many). Financial data is extracted from many Pdfs (1-Many) and the User can view many types of extracted financial data (1-Many). At a given point of time, when generating an answer, a chatbot only processes one piece of financial data (1-1). The User can ask multiple queries from a chatbot (1-Many) which provides many sentiment analysis (1-Many). The sentiment analysis is a weak entity due to its dependency on the chatbot and therefore the relationship between the chatbot and sentiment analysis is an identifying relationship. Sentiment Analysis is performed on a single piece of extracted financial data at a time (1-1) which is also an identifying relationship.

3.3 Data Design

The information domain of FinXtract is centered on financial data extracted from PDF reports and its transformation into data structures involves the following steps:

Data Storage:

MongoDB is used to store the financial data extracted from PDF reports in JSON format including the standardized and textual data.

FAISS vector database is used to store embeddings of data extracted from PDFs. These embeddings are representations of the extracted data in vector form used for similarity queries while interacting with the chatbot.

Backend Processing:

Node.js handles all the core backend processes which involve extracting data from PDFs, managing the API for user interactions and communicating with the MongoDB and the FAISS vector database for data retrieval.

Node.js uses the Ollama server to handle interaction with the chatbot, performing similarity searches across stored embeddings to retrieve the most relevant financial data for efficient query retrieval and utilizing natural language processing techniques such as sentiment analysis.